MERIDIAN PATHOLOGY LABORATORY

Clinical Pathology | Haematology | Biochemistry | Immunoassay
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REG. NO. DEMO/TS/PL/0001
(fictional laboratory)

LABORATORY INVESTIGATION REPORT

Patient Name	: **Ms. ANANYA DEMO (FICTIONAL)**	Lab ID	: 260914-0372
Age / Sex	: 38 Years / Female	Sample Collected	: 14/09/2026 07:45 AM
Patient ID	: MPL-2026-004817	Sample Received	: 14/09/2026 08:20 AM
Referred By	: Dr. S. Rao, MD (General Medicine)	Report Released	: 14/09/2026 04:10 PM

HAEMATOLOGY - COMPLETE BLOOD COUNT (CBC)

Specimen: Whole blood, EDTA

TEST	RESULT	FLAG	UNIT	REFERENCE INTERVAL	METHOD
Haemoglobin (Hb)	**11.8**	**L**	g/dL	12.0 - 15.5	*Photometry*
Total RBC Count	4.31		million/cumm	3.80 - 4.80	*Impedance*
Packed Cell Volume (PCV)	**35.9**	**L**	%	36.0 - 46.0	*Calculated*
Mean Corpuscular Volume (MCV)	83.3		fL	80.0 - 100.0	*Calculated*
Mean Corpuscular Hb (MCH)	27.4		pg	27.0 - 32.0	*Calculated*
Mean Corpuscular Hb Conc. (MCHC)	32.9		g/dL	31.5 - 34.5	*Calculated*
Red Cell Distribution Width (RDW-CV)	**14.9**	**H**	%	11.6 - 14.0	*Calculated*
Total Leucocyte Count (TLC)	7,450		cells/cumm	4,000 - 11,000	*Impedance*
Differential Leucocyte Count (DLC)					
Neutrophils	58		%	40 - 75	*Flow cytometry*
Lymphocytes	33		%	20 - 45	*Flow cytometry*
Eosinophils	6		%	1 - 6	*Flow cytometry*
Monocytes	3		%	2 - 10	*Flow cytometry*
Basophils	0		%	0 - 1	*Flow cytometry*
Absolute Neutrophil Count	4,321		cells/cumm	2,000 - 7,000	*Calculated*
Absolute Lymphocyte Count	2,459		cells/cumm	1,000 - 3,000	*Calculated*
Platelet Count	2.74		lakhs/cumm	1.50 - 4.10	*Impedance*
Mean Platelet Volume (MPV)	9.8		fL	7.5 - 11.5	*Calculated*
Erythrocyte Sedimentation Rate (ESR)	**28**	**H**	mm/1st hr	0 - 20	*Westergren*

Interpretation / Comments

Mild reduction in haemoglobin and PCV with raised RDW. Clinical correlation and iron studies are suggested.
ESR is a non-specific marker of inflammation; interpret together with clinical findings.

BIOCHEMISTRY - GLYCAEMIC STATUS

Specimen: Plasma (fluoride) / Whole blood EDTA

TEST	RESULT	FLAG	UNIT	REFERENCE INTERVAL	METHOD
Fasting Plasma Glucose (FPG)	**108**	**H**	mg/dL	70 - 100	*Hexokinase*
Post-Prandial Plasma Glucose (PPG)	**142**	**H**	mg/dL	70 - 140	*Hexokinase*
Glycosylated Haemoglobin (HbA1c)	**6.1**	**H**	%	4.0 - 5.6	*HPLC*
Estimated Average Glucose (eAG)	128		mg/dL	-	*Calculated*

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Interpretation / Comments - Glycaemic Status (continued)

HbA1c reference as per the laboratory: Normal < 5.7 %, Pre-diabetes 5.7 - 6.4 %, Diabetes >= 6.5 %.
Diagnosis of diabetes should not be made on a single laboratory value alone.

BIOCHEMISTRY - LIPID PROFILE

Specimen: Serum, 12-hour fasting

TEST	RESULT	FLAG	UNIT	REFERENCE INTERVAL	METHOD
Total Cholesterol	**214**	**H**	mg/dL	< 200	*CHOD-PAP*
Triglycerides	**168**	**H**	mg/dL	< 150	*GPO-PAP*
HDL Cholesterol	**46**	**L**	mg/dL	> 50	*Direct enzymatic*
LDL Cholesterol (Direct)	**134**	**H**	mg/dL	< 100	*Direct enzymatic*
VLDL Cholesterol	**33.6**	**H**	mg/dL	< 30	*Calculated*
Non-HDL Cholesterol	**168**	**H**	mg/dL	< 130	*Calculated*
Total Cholesterol / HDL Ratio	4.7			< 5.0	*Calculated*
LDL / HDL Ratio	2.9			< 3.5	*Calculated*

Interpretation / Comments

Target values depend on individual cardiovascular risk. Please discuss with your physician.

BIOCHEMISTRY - LIVER FUNCTION TEST (LFT)

Specimen: Serum

TEST	RESULT	FLAG	UNIT	REFERENCE INTERVAL	METHOD
Total Bilirubin	0.7		mg/dL	0.3 - 1.2	*Diazo*
Direct Bilirubin	0.2		mg/dL	0.0 - 0.3	*Diazo*
Indirect Bilirubin	0.5		mg/dL	0.1 - 1.0	*Calculated*
AST (SGOT)	31		U/L	5 - 34	*IFCC*
ALT (SGPT)	**38**	**H**	U/L	0 - 35	*IFCC*
Alkaline Phosphatase (ALP)	96		U/L	40 - 129	*IFCC*
Gamma GT (GGT)	24		U/L	6 - 42	*IFCC*
Total Protein	7.2		g/dL	6.4 - 8.3	*Biuret*
Albumin	4.1		g/dL	3.5 - 5.2	*BCG*
Globulin	3.1		g/dL	2.0 - 3.5	*Calculated*
Albumin / Globulin Ratio	1.3			1.0 - 2.1	*Calculated*

BIOCHEMISTRY - KIDNEY FUNCTION TEST (KFT)

Specimen: Serum

TEST	RESULT	FLAG	UNIT	REFERENCE INTERVAL	METHOD
Blood Urea	24		mg/dL	17 - 43	*Urease-GLDH*
Serum Creatinine	0.78		mg/dL	0.55 - 1.02	*Jaffe kinetic*
eGFR (CKD-EPI 2021)	98		mL/min/1.73 m2	> 90	*Calculated*
Uric Acid	5.4		mg/dL	2.6 - 6.0	*Uricase*

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BIOCHEMISTRY - MINERALS AND ELECTROLYTES

Specimen: Serum

TEST	RESULT	FLAG	UNIT	REFERENCE INTERVAL	METHOD
Calcium, Total	9.3		mg/dL	8.6 - 10.0	*Arsenazo III*
Phosphorus	3.6		mg/dL	2.5 - 4.5	*Molybdate*
Sodium (Na+)	139		mmol/L	136 - 145	*ISE indirect*
Potassium (K+)	4.3		mmol/L	3.5 - 5.1	*ISE indirect*
Chloride (Cl-)	103		mmol/L	98 - 107	*ISE indirect*

IRON STUDIES

Specimen: Serum

TEST	RESULT	FLAG	UNIT	REFERENCE INTERVAL	METHOD
Serum Iron	**48**	**L**	ug/dL	50 - 170	*Ferrozine*
Total Iron Binding Capacity (TIBC)	412		ug/dL	250 - 425	*Ferrozine*
Transferrin Saturation	**11.7**	**L**	%	15 - 50	*Calculated*
Serum Ferritin	**11**	**L**	ng/mL	13 - 150	*CLIA*

Interpretation / Comments

Low iron and ferritin with reduced transferrin saturation are consistent with depleted iron stores.

IMMUNOASSAY - THYROID PROFILE

Specimen: Serum

TEST	RESULT	FLAG	UNIT	REFERENCE INTERVAL	METHOD
Thyroid Stimulating Hormone (TSH)	**5.62**	**H**	uIU/mL	0.35 - 4.94	*CLIA*
Free Triiodothyronine (Free T3)	3.1		pg/mL	2.0 - 4.4	*CLIA*
Free Thyroxine (Free T4)	1.05		ng/dL	0.70 - 1.48	*CLIA*

IMMUNOASSAY - VITAMINS

Specimen: Serum

TEST	RESULT	FLAG	UNIT	REFERENCE INTERVAL	METHOD
25-OH Vitamin D (Total)	**18.4**	**L**	ng/mL	30.0 - 100.0	*CLIA*
Vitamin B12 (Cyanocobalamin)	**198**	**L**	pg/mL	211 - 911	*CLIA*

Interpretation / Comments

Vitamin D: Deficiency < 20 ng/mL, Insufficiency 20 - 29 ng/mL, Sufficiency 30 - 100 ng/mL.

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CLINICAL PATHOLOGY - URINE ROUTINE EXAMINATION

Specimen: Random urine, mid-stream

TEST	RESULT	FLAG	UNIT	REFERENCE INTERVAL	METHOD
Colour	Pale yellow			Pale yellow	*Visual*
Appearance	Clear			Clear	*Visual*
pH	6.0			5.0 - 8.0	*Reflectance*
Specific Gravity	1.015			1.005 - 1.030	*Reflectance*
Protein	Negative			Negative	*Reflectance*
Glucose	Negative			Negative	*Reflectance*
Ketones	Negative			Negative	*Reflectance*
Blood	Negative			Negative	*Reflectance*
Bilirubin	Negative			Negative	*Reflectance*
Urobilinogen	Normal			Normal	*Reflectance*
Nitrite	Negative			Negative	*Reflectance*
Leucocyte Esterase	Negative			Negative	*Reflectance*
Microscopic Examination					
Pus Cells	2 - 3		/HPF	0 - 5	*Microscopy*
Red Blood Cells	Nil		/HPF	Nil	*Microscopy*
Epithelial Cells	1 - 2		/HPF	Occasional	*Microscopy*
Casts / Crystals / Bacteria	Nil			Nil	*Microscopy*

***** End of Report *****

Medical Lab Technologist
Verified electronically

Dr. R. KAMALA DEMO, MD (Pathology)
Consultant Pathologist (fictional)

Conditions of reporting

1. Results relate only to the sample tested. 2. Reference intervals are specific to this laboratory and method.
3. Please correlate clinically; a single result should not be used in isolation to make a diagnosis.
4. This is a fictional document with invented data, created to test health-record software.

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